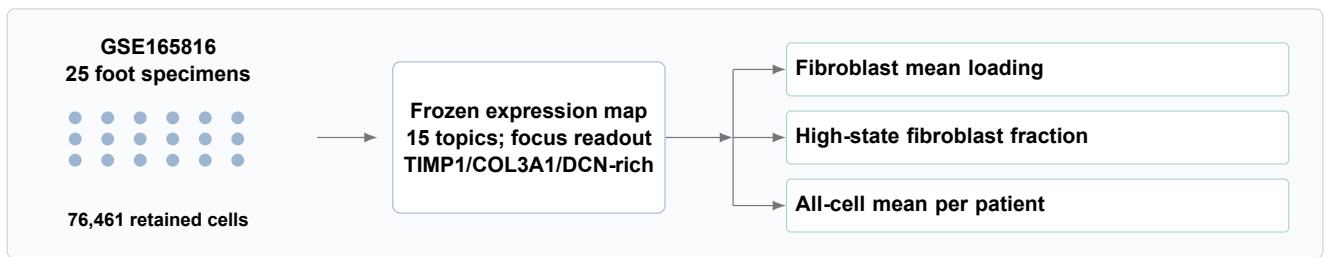
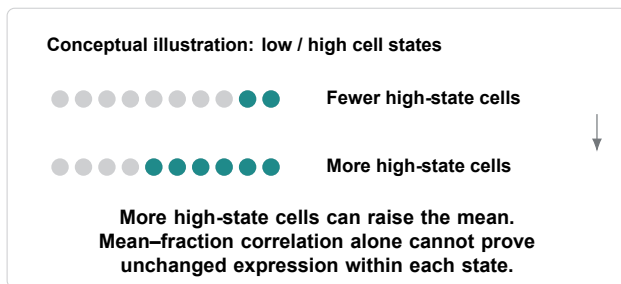


Figure 1

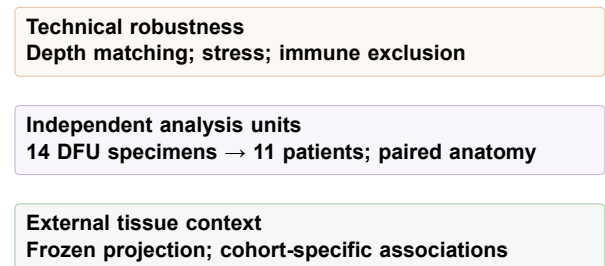
A From skin specimens to three distinct readouts



B What changes the average?



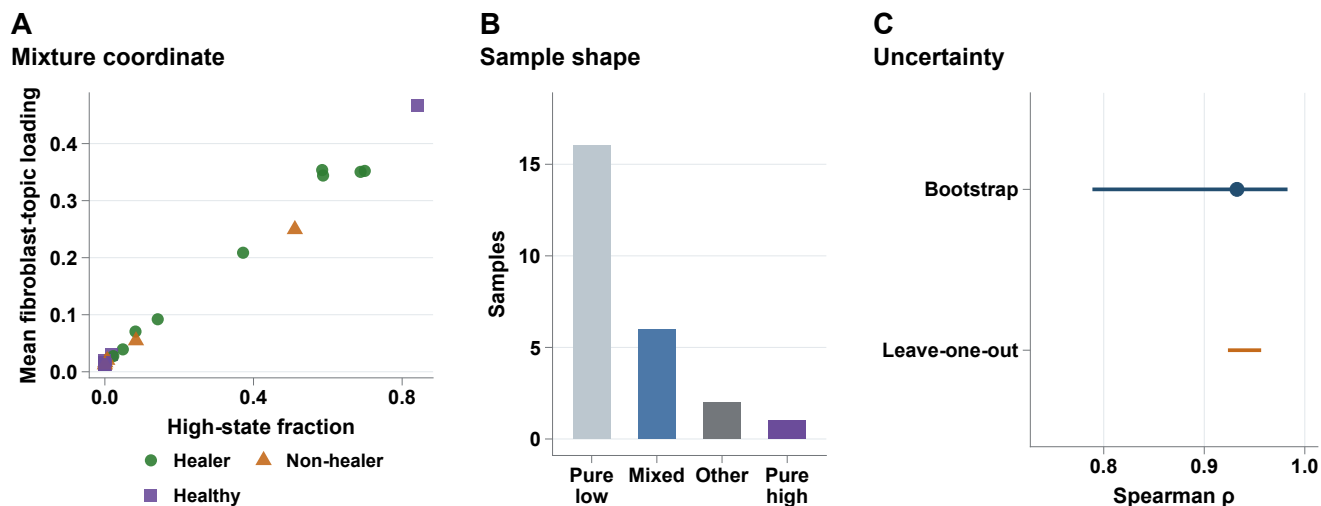
C How is the interpretation tested?



Cell-state abundance, technical robustness and healing prediction require separate evidence

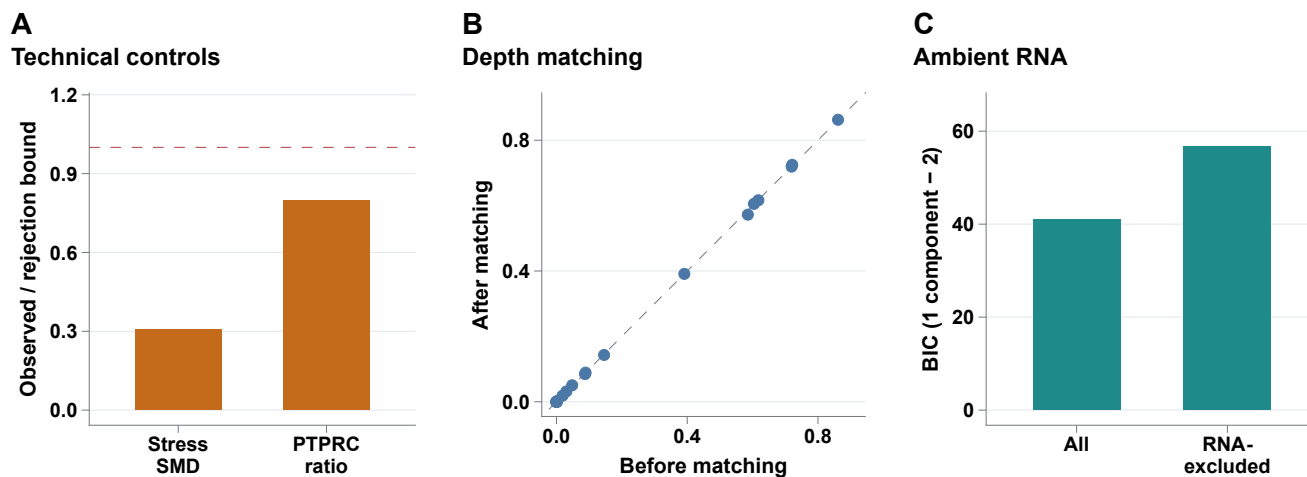
Study design and measurement logic. (A) The discovery series supplies 76,461 retained cells from 25 selected foot-skin specimens. A frozen 15-topic expression model produces three distinct summaries: the mean fibroblast loading, the fraction of fibroblasts above a fixed loading cut, and the all-cell mean within each mapped patient. The gene labels describe leading decoder genes, not a newly defined cell type. (B) The cell cartoons are conceptual, not observed counts: grey and teal circles denote low and high states. Changing their relative abundance can increase the sample mean without changing either state's intensity; a mean–fraction correlation alone cannot establish constant within-state expression. (C) Technical perturbations test robustness, the healing contrast collapses 14 DFU specimens into 11 patients, paired anatomy tests tissue sensitivity, and external projections assess cohort-specific construct associations. Each evidence layer addresses a different question; these analyses do not supply independent prognostic validation.

Figure 2



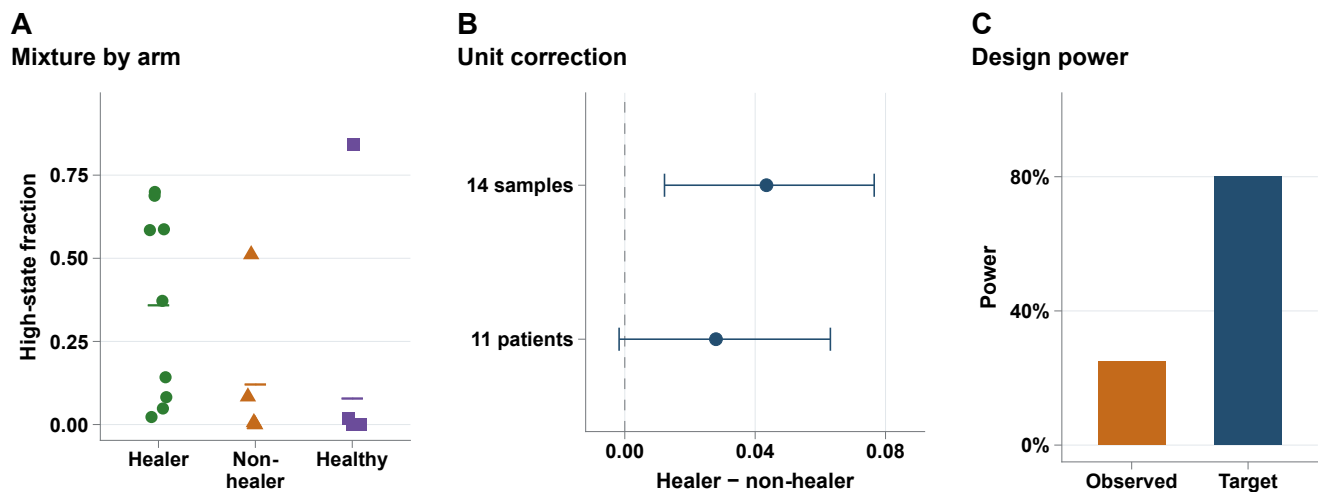
Association between mean topic loading within fibroblasts and the high-state fraction. (A) Each point is one of 25 GSE165816 specimens: 9 healing (green circles), 5 non-healing (orange triangles) and 11 healthy non-diabetic (purple squares). The high-state fraction uses the pooled cut 0.1842958; Spearman $\rho=0.933$. (B) Sample-shape counts use within-sample loading quantiles: pure low requires the 90th percentile at most 0.15, pure high the 10th percentile at least 0.10, and mixed the 10th percentile at most 0.05 and 90th percentile at least 0.40; other samples form the remaining category. (C) The dot is the observed correlation, the blue segment is the 10,000-draw sample-bootstrap 95% interval (0.789–0.983), and the orange segment is the leave-one-sample-out range (0.924–0.957). Both summaries use the same cells; the association does not by itself prove stable within-state intensity or a healing effect.

Figure 3

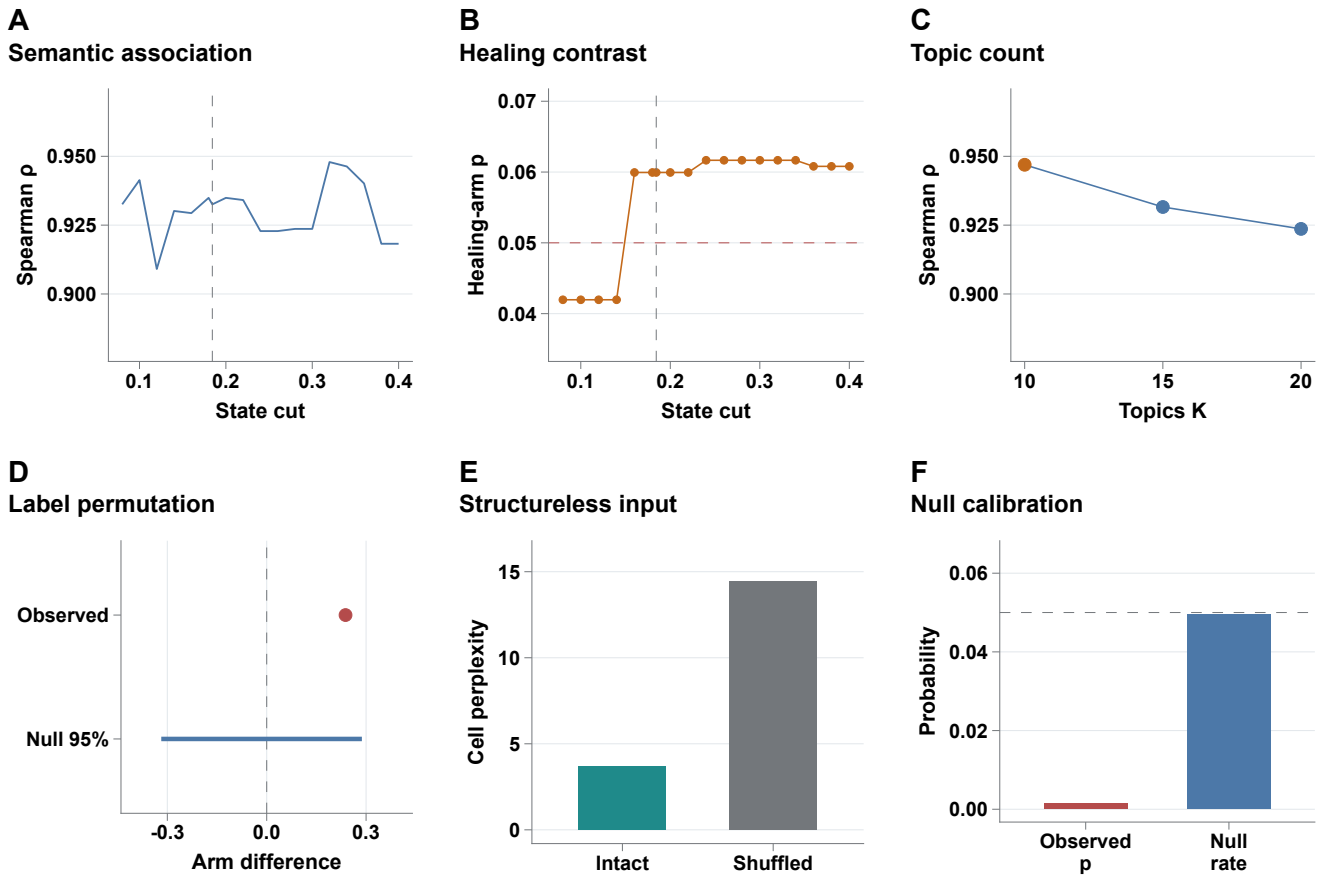


Technical and immune-transcript exclusion controls in the discovery cohort. (A) Dissociation standardized mean difference (SMD, 0.246) and the high-state/low-state PTPRC-positive rate ratio (1.60) are divided by rejection bounds 0.8 and 2.0; the dashed line is one. The rounded state cut is 0.184. (B) Each point is one of 25 samples after restricting to 14,561 fibroblasts eligible for panel-count matching to 986 counts per cell. Reprojection preserves sample fractions ($\rho=0.984$) and 99.5% of cell labels; the dashed line is identity. (C) Bars are BIC for one Gaussian minus BIC for two Gaussians fitted to sample high-state fractions, using 24 matched samples before and after exclusion. Positive values favour two components. Exclusion retains 11,262 of 19,410 fibroblasts overall, and matched sample fractions correlate at $\rho=0.989$. This control measures invariance to the exclusion rule; it does not identify every source of ambient RNA or tissue-composition dependence.

Figure 4

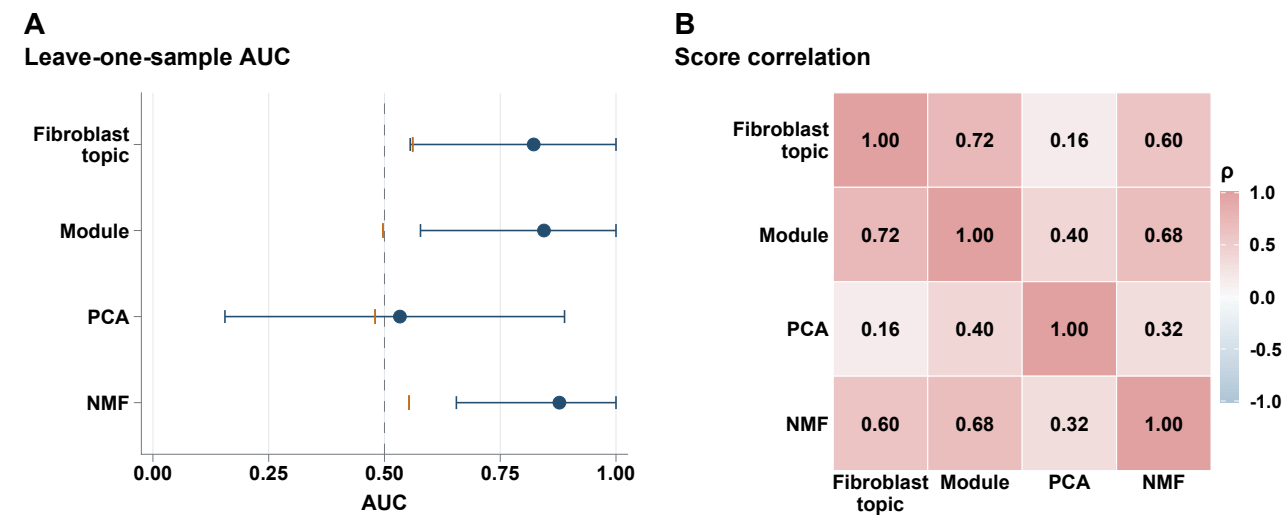


State abundance, analysis unit and design power. (A) Symbols show high-state fractions for 9 healing, 5 non-healing and 11 healthy specimens; horizontal segments mark arm means. Green circles, orange triangles and purple squares identify these groups. The largest fraction is 0.841 in healthy non-diabetic foot skin, showing that the state is not exclusive to healing ulcers. (B) Dots and within-arm bootstrap 95% intervals show the healing-minus-non-healing difference in all-cell mean fibroblast-associated topic loading for 14 specimens and 11 mapped patients. The patient analysis resamples 7 healing and 4 non-healing patients in 30,000 draws. Its readout differs from the fibroblast high-state fraction in A; the vertical dashed line is zero. (C) Gaussian location-shift simulation with a two-sided Mann–Whitney test estimates 24.9% power for the separate 9-versus-5 high-state-fraction design, against an 80% target. Its minimum detectable difference is 0.519. These are design sensitivities, not effect-exclusion bounds or clinical minimum important differences.

Figure 5

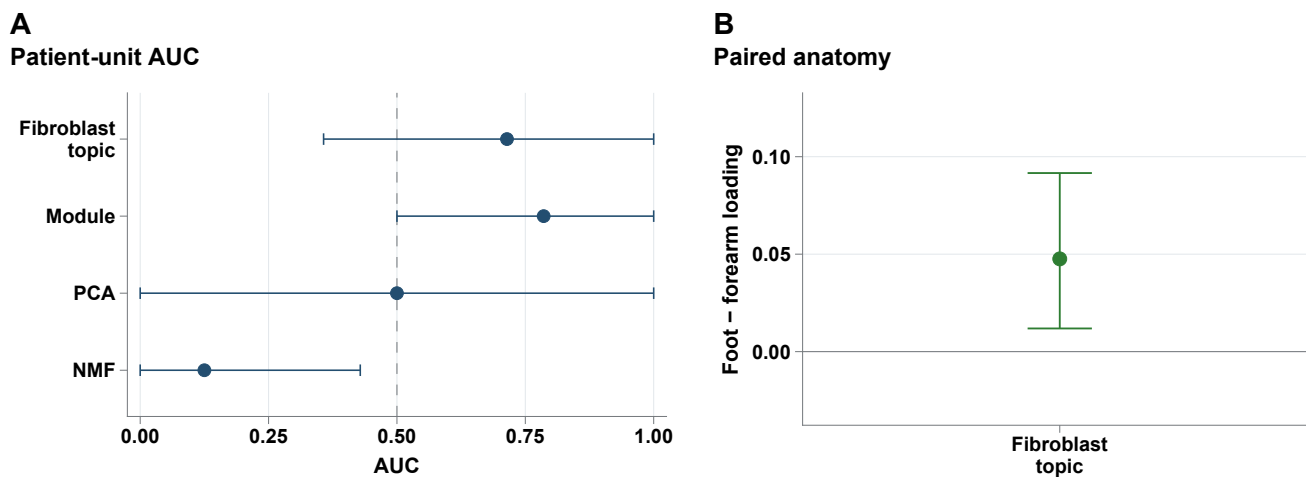
Threshold, capacity and negative-control analyses. (A,B) Spearman correlation of fibroblast mean loading with the high-state fraction across 25 samples and the two-sided Mann–Whitney healing-arm p value across 9 versus 5 samples, for all 18 cuts from 0.08 to 0.40. Vertical dashed lines mark the fitted cut; the horizontal line in B marks $p=0.05$. (C) Correlations for $K=10$, 15 and 20 after tracking the TIMP1-carrying topic; orange marks $K=10$, where the highest fraction is no longer in healthy skin. (D) The observed high-state-fraction arm difference is compared with the central 95% of 20,000 label permutations ($p=0.1358$). (E) Gene-column shuffling of 6,000 cells raises mean cell perplexity from 3.68 to 14.43 out of 15, violating the concentration criterion. (F) The anatomical sample-level control compares 33 foot and 11 forearm specimens: observed $p=0.0015$, with 5.0% of permuted comparisons below 0.05. This calibration is distinct from the paired 10-patient anatomy analysis and does not establish pipeline-wide error control.

Figure 6



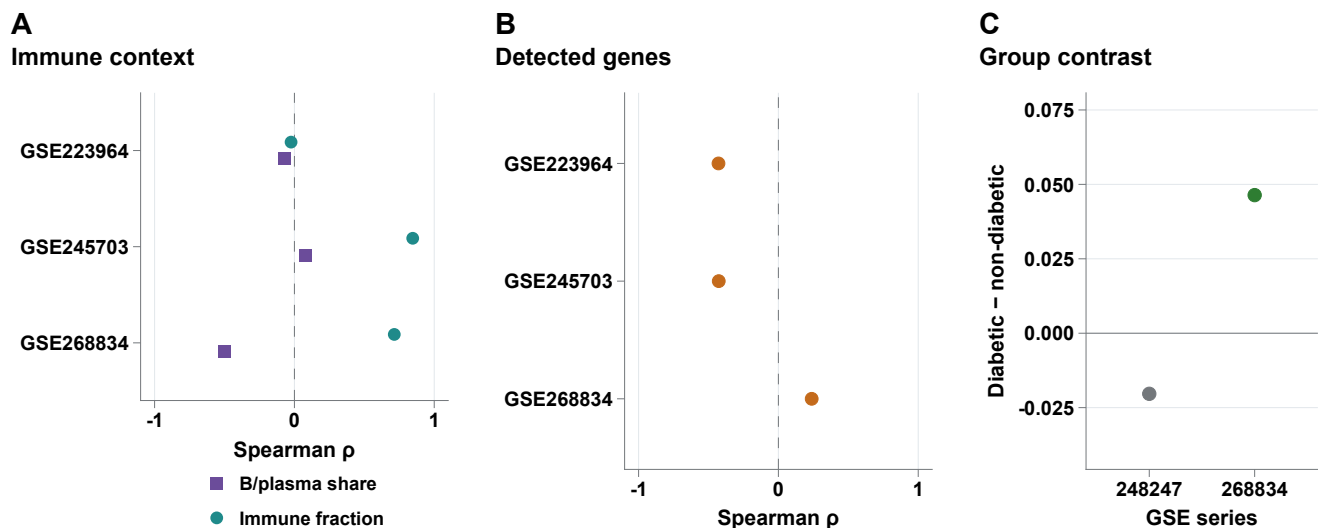
Sample-unit representation comparisons in 14 DFU specimens (9 healing, 5 non-healing). (A) Points are leave-one-sample-out areas under the ROC curve (AUCs); horizontal bars are percentile 95% intervals from 4,000 arm-stratified bootstrap draws of fixed out-of-fold scores, with common resampling indices across methods. Orange ticks mark the permutation-null means after rerunning outcome-dependent fitting and orientation. The dashed line at AUC=0.5 is a visual reference, not the fitted permutation null. Fibroblast topic denotes the frozen all-cell mean, Module the specified fibro-inflammatory score, PCA principal component analysis, and NMF nonnegative matrix factorization. No paired AUC-difference interval excludes zero. (B) Cells contain Spearman correlations of the 14 sample-level readouts; the colour scale and printed values encode the same statistic. These exploratory comparisons do not establish a clinical method ranking.

Figure 7



Patient-unit discrimination and paired anatomical sensitivity. (A) Points are leave-one-patient-out AUCs for 7 healing and 4 non-healing patients after cell-count-weighted collapse of repeated specimens. Bars are 95% intervals from 4,000 arm-stratified patient bootstrap draws of fixed out-of-fold scores; the dashed line marks AUC=0.5. Fibroblast topic, Module, PCA and NMF denote the frozen topic mean, fibro-inflammatory module, principal component and non-negative matrix factorization readouts. (B) The point is the mean all-cell fibroblast-associated topic loading difference, foot minus forearm, across 10 paired patients; the bar is a 4,000-draw paired-patient bootstrap 95% interval and the horizontal line is zero. The exhaustive sign-flip test with add-one correction gives $p=0.0107$ and Benjamini–Hochberg $q=0.0402$ across 15 topics. Both panels use the discovery source and are not independent clinical validation.

Figure 8



Frozen external projections under mutually exclusive lineage gates. (A) Each point is a cohort-level Spearman correlation between the sample fibroblast fibroblast-associated topic mean and either immune fraction among all QC cells (teal circles) or B/plasma share within immune cells (purple squares). (B) Orange points correlate the same topic mean with median detected genes per cell. GSE223964, GSE245703 and GSE268834 contribute 8, 12 and 8 specimens; vertical dashed lines mark zero. (C) Points show diabetic-minus-non-diabetic differences in the sample fibroblast mean: +0.0464 in GSE268834 (descriptive permutation $p=0.2812$) and -0.0204 in GSE248247 ($p=1.0$). GSE248247 contains one specimen per group and 358 QC cells, so its correlations are not estimated. No panel displays confidence intervals. These cohorts lack qualifying patient-level healing endpoints and are not pooled for prognosis inference.